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PROFILE

AI B.Sc. student (5th semester, GPA 2.3) with hands-on experience in Python, scikit-learn and time-series modeling from a research assistant role at THD. Developed full-stack ML and AI applications, including a voice-first help platform (HackaTUM 2025), an image classification pipeline, and a full-stack Titanic survival prediction app. Looking for an internship (min. 18 weeks) or working student position in Data Science, Machine Learning or Software Engineering from 02/2026 (20 h/week), in Germany (remote or on-site).

EDUCATION

- **Deggendorf Institute of Technology** *10/2023 - Present*
  Artificial Intelligence, B. Sc. Deggendorf, Germany
  - GPA: 2.3 (5th Semester); focus on Machine Learning, Software Engineering, Python Programming, and NLP
- **Vilnius University of Applied Sciences** *09/2025 – 11/2025*
  Erasmus+ Blended Intensive Programme — Student Management System Vilnius, Lithuania
  - Selected for an Erasmus+ programme to design and prototype a student management system, focusing on database architecture, authentication modules, and academic information workflows using Oracle APEX.
- **Selçuk University** *2019–2023*
  International Business and Relations, B. A. Konya, Turkey
  - GPA: 3.64/4.00
  - Focus: Business Development, Diplomacy, Research, International Trade

RELEVANT EXPERIENCE

- **Deggendorf Institute of Technology** *10/2025 – Present*
  Student Research Assistant Deggendorf, Germany
  - Preprocessed and cleaned 1+ year of multi-sensor water-quality data (pH, DO, conductivity, temperature) in Python (pandas, NumPy, scikit-learn), using Z-score–based outlier detection and time-series interpolation to handle missing values.
  - Developed turbidity and nitrate prediction pipelines by merging DF Robot sensor streams with meteorological variables, engineering cyclical time features and benchmarking 10 regression models (Linear/Ridge/ElasticNet/Huber, Random Forest/ExtraTrees, Gradient Boosting) using MAE, RMSE, and R^2 on a future holdout window.
  - Achieved $R^2 = 0.52$ and **RMSE** ≈ 0.29 for turbidity on the most recent 20-day window by implementing recency-weighted training (tuning $w_{\min}$ and power) and validating with prediction and residual plots.
  - Documented and communicated findings via structured LaTeX (Overleaf) reports, summarizing experiments, plots, and model behavior for internal team discussions.
- **Statista GmbH** *09/2023 – 11/2023*
  Research Assistant Hamburg, Germany
  - Processed, structured, and quality-checked datasets for market and industry statistics using Excel and internal tools, ensuring accuracy and consistency.
  - Supported content editors in curating and publishing statistical data and visualizations for the Statista.com platform and wrote concise English summaries for a global audience.

OTHER EXPERIENCE

- **Deggendorf Institute of Technology** *03/2025 – 07/2025*
  Orientation and Integration Tutor for International Students Deggendorf, Germany
  - Guided and supported international AI students with academic and administrative matters in English and German, improving communication and coordination skills.
  - Helped organize onboarding events such as Orientation Weeks and "Taste the World", fostering integration and intercultural exchange.
- **Özgayd Automotive** *08/2019 – 07/2022*
  Market Researcher Konya, Turkey
  - Identified over 100 potential cylinder head importers in South America using corporate databases, trade fair research, and outbound email/phone campaigns.
  - Contributed to an 80% increase in regional sales (2019–2022) through systematic lead generation, follow-up, and preparation of Excel-based customer lists and reports.

Academic Projects

- **ZeitZuHelfen — Voice-First Help Platform for Seniors** *2025*
  Tools: FastAPI, Telegram Bot API, Google Gemini 2.0 Flash, SQLite, SQLAlchemy, Pydantic
  - Developed during HackaTUM 2025: a voice-first help request platform connecting seniors in Munich with student helpers, reducing technology barriers for older adults.
  - Implemented a FastAPI backend with user management, authentication, and request CRUD operations, backed by a shared SQLite database using SQLAlchemy and async queries.
  - Built a Telegram bot interface that processes voice messages, using Google Gemini 2.0 Flash to transcribe speech and extract structured help-request data.
  - Integrated Google Address Validation API and gTTS to validate German addresses and provide natural German voice responses to users.
- **Stone & Green-Area Detection (Embedded Object Detection)** *2025*
  Tools: Python, OpenCV, YOLOv4, LabelImg, Raspberry Pi
  - Built an embedded computer-vision pipeline to detect stones and classify green areas for a small-scale outdoor navigation scenario.
  - Collected and **manually annotated** images with bounding boxes (YOLO format), maintaining a clean dataset structure and train/validation split.
  - Trained and fine-tuned a **YOLOv4** object detector; improved robustness using **data augmentation** (flips, rotations, brightness/contrast, and scaling).
  - Deployed and tested the model on a **Raspberry Pi** setup, validating real-time inference behavior and iterating on data quality to reduce false positives.
- **IcebergAI — Full-Stack ML Web Application** *2025*
  Tools: Python, scikit-learn, React+Vite, Docker, GitLab
  - Group software engineering project; built an end-to-end ML system to predict Titanic passenger survival based on demographic and ticket features.
  - Implemented and evaluated classification models in Python (scikit-learn) and exposed the prediction logic via a backend API.
  - Designed a modern, responsive React+Vite frontend to collect user input and display prediction results in an intuitive way.
  - Containerized the application with Docker and coordinated a 7-member team using GitLab issues and merge requests to ensure on-time delivery.

Technical Skills

- **Programming & Data:** Python, SQL, pandas, NumPy
- **ML & AI:** scikit-learn, PyTorch, LangChain, time-series modeling, model evaluation (MSE, MAE, R^2)
- **Web & APIs:** FastAPI, Flask, Streamlit, React, Telegram Bot API, Oracle APEX
- **Databases & BI:** MySQL, PostgreSQL, SQLite, Power BI
- **Tools & DevOps:** Git, GitHub/GitLab, Docker, Linux, Jupyter, Google Colab, LaTeX, Grafana

Awards

- **HackaTUM 2025 – City of Munich Challenge** *11/2025*
  TUM Munich
  - Placed in the Top 10 teams with the *ZeitZuHelfen* project.
- **STIBET Scholarship** *2025*
  DAAD / THD
  - Awarded for outstanding commitment at Deggendorf Institute of Technology.
- **Erasmus+ Scholarship** *2025*
  Erasmus+
  - Funded participation in an Erasmus+ Blended Intensive Programme in Vilnius.

Training and Certifications

- LangChain – Develop AI Agents with LangChain and LangGraph (Udemy)
- Supervised Machine Learning: Regression and Classification (Coursera)

Languages

Spanish (Native), English (C1), German (B2), Turkish (C1), Portuguese (A1)